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3.3 Study of Proposed System The proposed system combines federated learning with explainable artificial intelligence for wafer defect detection. Rather than relying on centralized data pooling that forces semiconductor foundries to expose proprietary fabrication data, the framework trains a global convolutional neural network across distributed clients without moving raw wafer images. The framework uses data collected through the MixedWM38 wafer map benchmark comprising over 38,000 grayscale wafer images across 38 fault classes. The study examines wafer spatial patterns Department of Computer Science and Design,FISAT 10Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection characteristics such as center, donut, edge-ring, and mixed-type defect clusters. These characteristics are used to assess whether individual dies across the wafer satisfy strict semiconductor manufacturing yield standards. The collected wafer information is incorporated into the federated aggregation process. This enables the central model to learn robust representations across non-IID client distributions while preserving facility data confidentiality. The model first trains localized convolutional feature extractors across ten manufacturing nodes. When local client epochs conclude, the system applies Federated Averaging with calibrated Gaussian noise injection to provide differential privacy guarantees. The paper also presents an explainable AI layer combining Grad-CAM and LIME to support the transparent interpretation of predictions. The application is intended to communicate feature attribution heatmaps directly to foundry process engineers, making defect detection trustworthy and actionable. The overall approach connects decentralized training, privacy-preserving aggregation, and visual interpretability in a single framework. 3.4 Requirement Specification The proposed semiconductor fault detection framework involves wafer image ingestion, local client model training, secure gradient aggregation, and post-hoc visual explanation generation. Therefore, suitable data, processing capabilities, and software components are required to support the workflow. The study depends on the availability of standardized wafer map data derived from multi-stage fabrication sensors. The collected data must describe thirty-eight distinct single and mixed-type wafer defect patterns. A suitable computational environment is also required to process deep convolutional backbones and calculate differential privacy budgets. 3.4.1 Recommended System Requirements The following requirements describe the general computational resources needed to support a federated wafer inspection framework. They should not be interpreted as exact hardware specifications reported by the paper: • Processor: A high-performance multi-core CPU coupled with dedicated GPU acceleration (NVIDIA RTX 3080 or equivalent) to handle parallel convolutional tensor calculations. • Memory: Sufficient RAM to store normalized 52x52 wafer batch arrays and gradient update caches (Minimum 16GB, 32GB recommended). • Storage: Adequate high-speed SSD storage for caching the MixedWM38 dataset, model checkpoints, and t-SNE clustering projections. • Network Connectivity: Low-latency local network or TLS-encrypted tunnel required for transferring weight update tensors between client inspection nodes and the central aggregation server. • Execution Environment: Linux-based operating system (Ubuntu 22.04 LTS) configured with CUDA acceleration libraries for client training. Department of Computer Science and Design,FISAT 11Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection • Explainability Module: Dedicated compute capacity to generate pixel-level Grad-CAM heatmaps and LIME super-pixel perturbation surrogates for real-time defect verification. 3.4.2 Software Requirements The framework requires software components capable of training deep residual networks, managing decentralized communication rounds, injecting differential privacy noise, and generating visual heatmaps: • Data Preprocessing: Image processing libraries for grayscale normalization, pixel standardization, and one-hot encoding of 38 defect categories. • Machine Learning Engine: PyTorch / TensorFlow runtime supporting ResNet152, DenseNet121, MobileNetV2, and InceptionV3 backbones. • Federated Orchestration: Framework supporting Federated Averaging (FedAvg) and client node scheduling across 10 communication rounds. • Privacy Engine: Differential privacy routines for calibrated Gaussian perturbation of global weight updates. • Explainable AI Suite: Grad-CAM gradient backpropagation modules and LIME local surrogate algorithms for interpretable inspection. 3.5 Functional Requirements The federated wafer fault detection framework should satisfy the following functional requirements: 1. The system shall partition the wafer map dataset into ten private client shards representing decentralized fabrication stations. 2. The system shall train local convolutional models on private wafer image data without exporting raw wafer images off-site. 3. The system shall evaluate four deep learning architectures (ResNet152, DenseNet121, MobileNetV2, InceptionV3) using residual skip connections. 4. The system shall combine client weight updates at a central coordinator server using Federated Averaging (FedAvg). 5. The system shall inject calibrated Gaussian noise into aggregated model weights to guarantee formal differential privacy. 6. The system shall achieve high classification accuracy across all thirty-eight single, two-mixed, three-mixed, and four-mixed defect categories. 7. The system shall generate Grad-CAM class activation heatmaps highlighting the exact die regions responsible for defect classification. Department of Computer Science and Design,FISAT 12Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection 8. The system shall construct LIME super-pixel segmentation overlays to corroborate model focus. 9. The system shall support feature space visualization via t-SNE clustering to verify cross-client latent representation quality. 10. The system shall support the presentation of automated defect inspection

recommendations through an intuitive inspection dashboard for fab engineers. 3.6 Non-Functional Requirements The federated semiconductor inspection framework should satisfy the following non-functional requirements: • Privacy Preservation: Raw wafer images must strictly reside within client fabrication boundaries, and differential privacy noise must prevent gradient reconstruction attacks. • Accuracy and Reliability: The global federated classifier must achieve high generalization accuracy, matching or exceeding centralized baselines (reaching a verified 98.78% test accuracy with ResNet152, outperforming DCNet by 5.58 percentage points). • Convergence and Communication Efficiency: The federated optimization protocol must achieve stable convergence within a constrained budget of 10 communication rounds. • Transparency and Explainability: Model predictions must be accompanied by faithful, human-interpretable visual explanations (Grad-CAM and LIME) to eliminate black-box opacity and establish domain trust among foundry engineers. • Robustness to Data Heterogeneity: The framework must maintain stable classification performance across unbalanced defect distributions and varying client dataset volumes. • Scalability: The architecture must easily scale to accommodate additional manufacturing facilities and new wafer geometry formats without requiring architectural reconfiguration. • Maintainability: Modular separation between the client training nodes, aggregation server, and XAI interpretation pipeline must allow independent algorithmic updates. 3.7 Summary This chapter analysed the system overview, architectural design, dataset characteristics, and technical requirements of the federated learning and explainable AI framework for semiconductor defect detection. The reviewed framework addresses the privacy bottlenecks of centralized training and the black-box opacity of deep neural networks by combining decentralized FedAvg, differential privacy, and dual XAI interpretability layers. The proposed approach connects private localized training on 38,000 wafer maps with transparent attribution heatmaps, demonstrating superior accuracy and operational trustworthiness for multi-stakeholder industrial semiconductor manufacturing. Department of Computer Science and Design,FISAT